Chapter 1: Numerical Methods

In this chapter, we introduce a locally mass conserving numerical framework. A conforming mixed finite element discretization (MFEM) is proposed to solve the static Darcy-Stokes coupled methcnics system. The proposed conforming MFEM scheme captures the normal fluxes across element edges, ensuring compatibility with local mass conservation modification. The transport system is solved using a finite volume method (FVM), which inherently ensures local mass conservation.

We begin by introducing the mesh partition used for both the MFEM and the FVM schemes. Based on the mesh partition, two stable element pairs are presented, one for Darcy flow and one for Stokes flow, each equipped with the direct serendipity space. Subsequently, the entire coupled system is discretized with the previously defined finite element pairs. We then present a noval nonlinear WENO weighting scheme for the FVM. Error estimatea are provided for both the MFEM and the FVM schemes.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$, where $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. For a given $h > 0$, let $\mathcal{T}_h$ be a *triangulation* of the closure of the domain $\overline{\Omega}$ made of quadrilaterals E, i.e.,

$$\overline{\Omega} = \cup_{E\in\mathcal{T}_h} E, \quad h_E < h, \tag{1.1}$$

where two quadrilaterals are either disjoint or sharing one edge or one vertex. We assume that the quadrilateral mesh is uniformly shape regular. A formal definition of uniform shape regularity of a quadrilateral mesh is given in Girault and Raviart (2011).

Definition 1.1. For any quadrilateral $E \in \mathcal{T}_h$, we can get a set of four triangles $T_i, \quad i \in \{0, 1, 2, 3\}$. We form the triangle T_i by excluding vertex v_i from the quadrilateral E, and use the remaining three vertex. We define the following parameters as

$$\begin{aligned} &h_E = \text{diameter of E}, \\ &\rho_E = 2 \min_{1 \leq i \leq 4} \{\text{diameter of largest circle inscribed in } T_i\}. \end{aligned} \tag{1.2}$$

A mesh partition $\mathcal{T}_h$ is considered to be uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that for all quadrilaterals $E \in \mathcal{T}_h$ we have the ratio lower bound as

$$\rho_E / h_E \geq \sigma_* > 0, \tag{1.3}$$

where no subtriangle T_i of E has zero measure, i.e., each quadrilateral E is assumed to be strictly convex, not degenerate into triangle, line or point.

The quadrilateral $E \in \mathcal{T}_h$ is referred to differently depending on the contexts. In finite element method, a quadrilateral E is being called as an *element*, while in finite volume method, it is typically referred to as a *cell*. For simplicity, we use the term *element* throughout this section about mesh.

In Figure 1.1, we show a reference element $\hat{E} = [-1, 1]^2$, a local reference element $\tilde{E}$ and a general quadrilateral element E. The edges of element E are denoted by e_i, $i \in \{0, 1, 2, 3\}$, and the vertices of the element E are denoted as v_i, $i \in \{0, 1, 2, 3\}$ both numbered in the counterclockwise order. We begin counting from zero, first to align with the indexing convention of C++, and second to facilitate representation in terms of modulo arithmetic, which also starts from zero. The vertices can be understood by $v_i = e_i \cap e_{i+1}$, where $i \equiv i \pmod 4$. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outwards from the element E. The corresponding unit tangent $\boldsymbol{\tau}_i$ is obtained by rotating $\boldsymbol{\nu}_i$ counterclockwise, following the right-hand rule.

We assume the general quadrilateral element E can be obtained by scaling from the local reference element $\tilde{E}$ such that $\tilde{E} = E/H$, where H is the maximal edge length of E, i.e. , the length of edge e_1 in Figure 1.1. Here, $|e_1| = H$ and $|\tilde{e}_1| = 1$. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to \tilde{E}$ mapping the vertices of $\hat{E}$ to the vertices of $\tilde{E}$. Note that the unit normal and the unit tangent vectors are invariant under scaling mapping H.

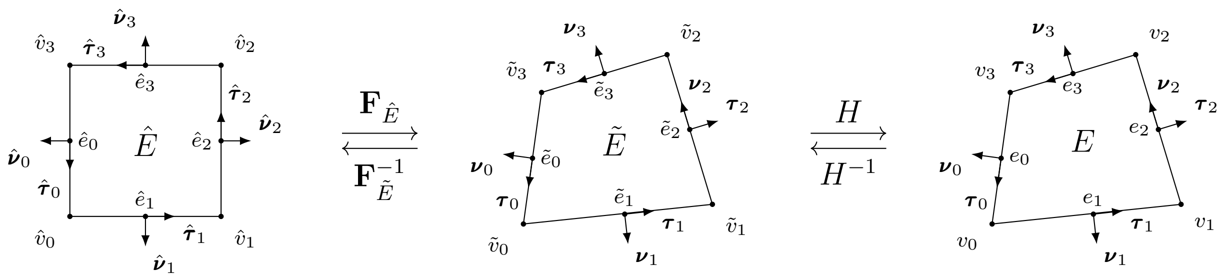


Figure 1.1: An exhibition of a reference square element, a local reference element, and a general non-degenerate quadrilateral element.

For the ease of implementation, we use a logically rectangular mesh, where each element is indexed using Cartesian coordinates $\{i, j\}$. In Figure 1.2, we illustrate a full logically rectangular, quasi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distorting the interior vertex of a square mesh. The boundary vertices $v_i \in \partial\Omega$ remain fixed and are not subject to distortion.

1.2 Mathematical Preliminaries

In this section, we introduce some mathematical concepts that are relied upon in the remainder sections. The theory and notations are presented in a general dimension sense, but in practical, our discussion are confined to two dimension cases. In addition, for simplicity, we restrict our consideration to real-valued functions.

To begin with, we recall the definition of a Sobolev space.

Definition 1.2. Let $\Omega \subset \mathbb{R}^d$ be a domain, $1 \le p \le \infty$, and $m \ge 0$ be an integer.

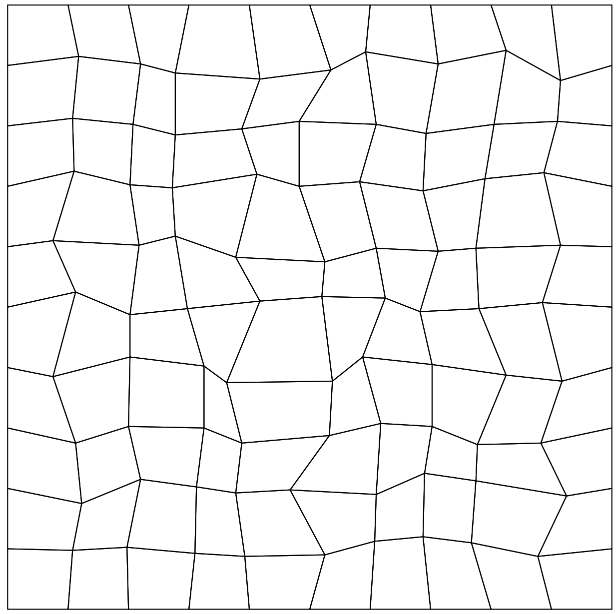


Figure 1.2: An exhibition of a full logically rectangular quasi-uniform quadrilateral mesh $\mathcal{T}_h$.

The Sobolev space of m derivatives in $L^p(\Omega)$ is

$$W_p^m(\Omega) = \{f \in L^p(\Omega) : D^\alpha f \in L^p(\Omega) \text{ for all multi-indices } \alpha \text{ such that } |\alpha| \leq m\}, \tag{1.4}$$

where the multi-indices $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_{N-1}) \in \mathbb{N}^N$ and

$$|\alpha| = \sum_{i=0}^{N-1} \alpha_i.$$

The multi-indices derivative is defined as

$$D^\alpha = \frac{\partial^{|\alpha|}}{\partial x_0^{\alpha_0} \ldots \partial x_{N-1}^{\alpha_{N-1}}}.$$

For $f \in L^p(\Omega)$, the $L^p(\Omega)$ norm is

$$\|f\|_{L^p(\Omega)} = \left(\int_\Omega |f|^p \right)^{1/p}. \tag{1.5}$$

We proceed with the definition of the Sobolev norm and seminorm.

Definition 1.3. For $f \in W_p^m(\Omega)$, the norm $\|\cdot\|_{W_p^m}$ is

$$\|f\|_{W_p^m(\Omega)} = \left\{ \sum_{|\alpha| \leq m} \|D^\alpha f\|_{L^p(\Omega)}^p \right\}^{1/p}, \quad p < \infty, \tag{1.6}$$

and

$$\|f\|_{W_\infty^m(\Omega)} = \max_{|\alpha| \leq m} \|D^\alpha f\|_{L^\infty(\Omega)}, \quad p = \infty. \tag{1.7}$$

The seminorm $|\cdot|_{W_p^m}$ is

$$|f|_{W_p^m(\Omega)} = \left\{ \sum_{|\alpha| = m} \|D^\alpha f\|_{L^p(\Omega)}^p \right\}^{1/p}, \quad p < \infty. \tag{1.8}$$

In the special case when $p = 2$, we denote $W_2^m(\Omega) = H^m(\Omega)$. The corresponding norm and seminorn are then written as $\|\cdot\|_{H^m(\Omega)}$ and $|\cdot|_{H^m(\Omega)}$ respectively. Recall the Trace Theorem in $H^m(\mathbb{R}^d)$ space.

Theorem 1.1. *(Trace Theorem)*

Let k and d be integers with $0 < k < d$. The restriction map R extends to a bounded linear map from $H^m(\mathbb{R}^d)$ onto $H^{s-k/2}(\mathbb{R}^{d-k})$, provided that $s > k/2$.

Proof. See Arbogast and Bona (2025). □

Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ can be calculated as

$$\dim \mathbb{P}_r(\mathbb{R}^d) = \binom{r+d}{d} = \frac{(r+d)!}{r!d!}. \tag{1.9}$$

We refer to the Bramble-Hilbert lemma for basic interpolation error estimation for polynomials of degree r.

Lemma 1.2. *(Bramble-Hilbert Argument)*

Let $\Omega \in \mathbb{R}^d$ be a domain, and for function $f \in W_p^m(\Omega)$, $m \geq 0$, $1 \leq p < \infty$. There exists a constant C such that the inequality holds, i.e, that

$$\inf_{p \in \mathbb{P}_{k-1}(\Omega)} \|f - p\|_{W_p^m(\Omega)} \leq C h^{k-m} |f|_{W_p^k(\Omega)}, \quad j = 0, 1, \ldots, k, \tag{1.10}$$

where the constant C has an upper bound due to shape regularity assumption.

Proof. See Bramble and Hilbert (1970) and Dupont and Scott (1980). □

1.3 Direct Serendipity Space

Serendipity elements have the same accuracy of approximation with classical tensor product element while use fewer degrees of freedom. However, the classical serendipity space suffer from poor approximation on quadrilaterals. To address this issue, we follow the approach of Arbogast et al. (2022) by defining local shape functions directly on quadrilateral elements, thereby recovering the desired accuracy. In the next sections, H^1 and $H(\text{div})$ conforming direct mixed finite element space are constructed with the direct serendipity function space.

To begin with, we introduce some auxiliary distance functions. We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.11}$$

where $\mathbf{x}_i^* \in e_i$ is any point on edge e_i. We pick $\mathbf{x}_i^* = \mathbf{x}_{v,i}$ for simplicity, where $\mathbf{x}_{v,i}$ denotes the coordinates of corner v_i. The value of distance function λ_i returns a positive value as long as the point is located in the interior of the element E.

We define two additional distance functions giving the distance of a point at coordinate $\mathbf{x} \in \mathbb{R}^2$ to diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\boldsymbol{\nu}_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by rotating counterclockwise of the unit vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{d,i}^*) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 0, 1, \tag{1.12}$$

where $\mathbf{x}_{d,i}^*$ is any point on the diagonal line d_i. Here, we use the corner point v_i for simplicity, i.e., $\mathbf{x}_{d,i}^* = \mathbf{x}_{v,i}$.

We now present a set of nodal basis functions for the first and second order direct serendipity spaces, $\mathcal{DS}_1$ and $\mathcal{DS}_2$.

1.3.1 Second-order Space $\mathcal{DS}_2$

The dimension of second-order polynomial in two dimension is calculated by equation 1.9 as

$$\dim\mathbb{P}_2(\mathbb{R}^2) = \binom{2+2}{2} = \frac{(4)!}{2!2!} = 6, \tag{1.13}$$

while a quadrilateral element provide a minimum of 4 vertex dofs and 4 edge dofs. Therefore, we supplement polynomial space $\mathbb{P}_2(E)$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$, spanning the supplemental function space as $\mathbb{S}_r^{\mathcal{DS}}(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\},\ \ r \geq 2$. We present the second-order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.14}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{0,1\}, \tag{1.15}$$

where $i \equiv i \pmod 4$. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.16}$$

which satisfies the conditions as

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.17}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \quad i \in \{0,1,2,3\} \tag{1.18}$$

where $i \equiv i \pmod 4$ as usual, and $R_{i,i+2} = -R_{i+2,i}$ according to the equation (1.16). Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral element, which evaluates 1 on the edge e_0, and evaluates 0 on the opposite edge e_2.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{e,i}^{(2)}(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.19}$$

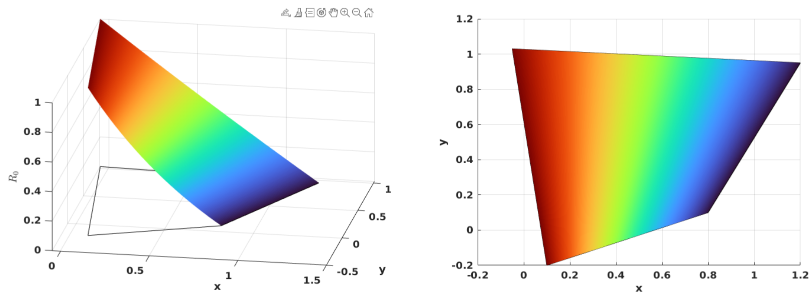


Figure 1.3: Rational function R_0.

where $\mathbf{x}_{e,i}$ is the coordinates of the middle point of the edge e_i, again $i \equiv i \pmod 4$. We use superscript (2) to denote the order of the function space. We can restrict the edge basis function to edges as

$$\varphi_{e,i}^{(2)}|_{e,j} = \begin{cases} \dfrac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})}, & i = j \\ 0, & i \neq j \end{cases} \tag{1.20}$$

In Figure 1.4, we present edge basis function $\varphi_{e,0}^{(2)}$ on a general quadrilateral element, where $\varphi_{e_0}^{(2)}$ evaluates 0 on the edges $\{e_i, \quad i \neq 0\}$, and evaluates a quadratic function on the edge e_0.

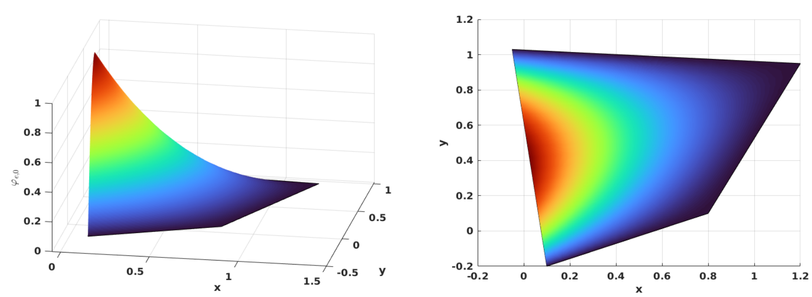


Figure 1.4: Edge nodal basis function $\varphi_{e,0}^{(2)}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{v,i}^{(2)}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{2}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}_{v,i})}, \tag{1.21}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i, and $i \equiv i \pmod 4$ as well. In Figure 1.5, we show $\varphi_{v,0}^{(2)}$ on a general quadrilateral.

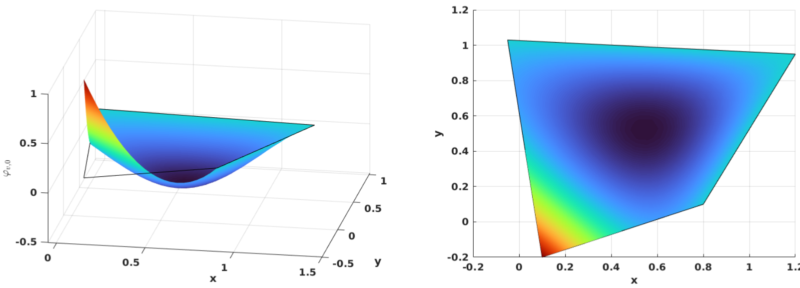


Figure 1.5: Vertex nodal basis function $\varphi_{v,0}^{2}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.3.2 First-order Space $\mathcal{DS}_1$

The dimension of first-order polynomial in two dimension is also calculated by equation 1.9 as

$$\dim\mathbb{P}_1(\mathbb{R}^2) = \binom{1+2}{2} = \frac{(3)!}{2!1!} = 3, \tag{1.22}$$

while a quadrilateral element provide a minimum of 4 vertex dofs. Therefore, we supplement polynomial space $\mathbb{P}_1(E)$ with one additional function. The supplemental function for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order direct serendipity space as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \mathrm{span}\{R\}, \tag{1.23}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of the rational function R is not unique. We present one way to construct $R(\mathbf{x})$ using edge nodal basis function (1.19) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi_{e,i}^{(2)}(\mathbf{x}), \tag{1.24}$$

where the auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.25}$$

In Figure 1.6, we plot the rational function $R(\mathbf{x})$ on a general quadrilateral.

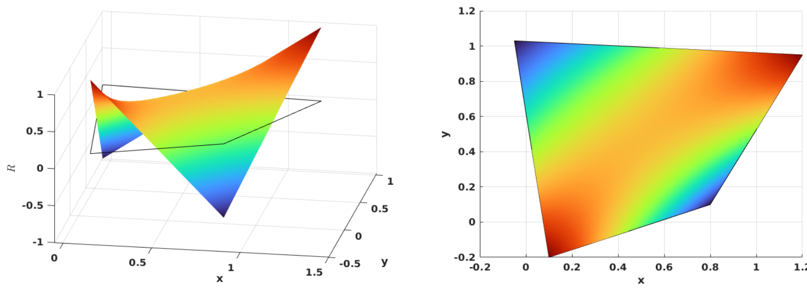


Figure 1.6: Supplemental rational function R in $\mathcal{DS}_1$.

We then define vertex nodal basis function accordingly as

$$\varphi_{v,i}^{(1)}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.26}$$

where $m \equiv i+1 \pmod 2$, and $i \equiv i \pmod 4$. In Figure 1.7, we show the vertex nodal basis function $\varphi_{v,0}^{(1)}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertices.

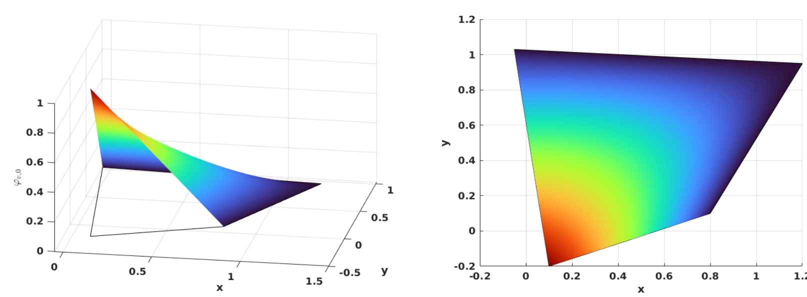


Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

1.3.3 An Interpolation Operator

For completeness, we construct an interpolation operator mapping onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$ following Arbogast et al. (2022). For each node a_i, the associated K_i has different meanings depending on the type of basis functions. If the basis function is associated with an edge node, we set $K_i = e_i$, where e_i is the corresponding edge. For the basis function associated with a vertex node, we set K_i to be any edge e containing node a_i, with the restriction that if $a_i \in \partial\Omega$, then $e \in \partial\Omega$.

An interpolation operator $\mathcal{I}_h^r : W_p^l(\Omega) \to \mathcal{DS}_r$ is defined as

$$\mathcal{I}_h^r v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y}) d\mathbf{y} \in \mathcal{DS}_r, \tag{1.27}$$

where $1 \le p \le \infty$ and $l > 1/p$ (but $l \ge 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x}) \varphi_j(\mathbf{x}) d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \dots, N_r\}, \tag{1.28}$$

where N_r is the number of nodal basis functions. We present the following two lemmas regarding the constructed interpolation operator for future reference.

Lemma 1.3. *(Boundedness of the interpolation operator.)*

Let $\mathcal{T}_h$ be uniformly shape regular according to Definition (1.3) with shape regularity parameter σ_*. For each element $E \in \mathcal{T}_h$, There exists a bounded interpolation operator (1.27). Let $v \in W_p^l(\Omega)$, with $1 \le p \le \infty$ and $l > q/p$. Suppose that the basis functions of $\mathcal{DS}_r(E)$ are constructed using λ_{24} and λ_{13} such that the zero set $\mathcal{Z}_{i,i+2}$ intersects e_{i+3} and e_{i+1}. Moreover, assume that $R_{i,i+2}$ is uniformly differentiable functions of the vertices of E up to order m. Then for $r \ge 1$, $E \in \mathcal{T}_h$, $1 \le q \le \infty$, and any non negative integer m,

$$\|\mathcal{I}_h^r v\|_{W_q^m(E)} \le C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{1.29}$$

where $E^* = \cup_{F \in \mathcal{T}_h, F \cap E \neq \emptyset} F$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives.

Proof. See Arbogast et al. (2022). □

Combining Lemma 1.3 and Bramble-Hilbert lemma 1.2, we give local and global error estimation.

Lemma 1.4. *(Approximability of the interpolation operator.)*

Under the assumption of Lemma 1.3, there exists a constant $C = C(r, \sigma_*) > 0$ *such that for all functions* $v \in W_{l,p}(E^*)$*, with* $1 \leq p \leq \infty$ *and* $l > 1/p$ *or (*$l \geq 1$ *if* $p = 1$*),*

$$\|v - \mathfrak{I}_h^r v\|_{W_p^m} \leq C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \leq m \leq \min(l, r+1). \tag{1.30}$$

Moreover, there exists a constant $C = C(r, \sigma_*) > 0$*, independent of* $h = \max_{E \in \mathfrak{T}}$*, such that for all functions* $v \in W_p^l(\Omega)$*,*

$$(\sum_{E \in \mathfrak{T}_h} \|v - \mathfrak{I}_h^r v\|_{W_p^m(E)}^p)^{1/p} \leq C h^{l-m} |v|_{W_p^l(\Omega)}, \quad 0 \leq m \leq \min(l, r+1). \tag{1.31}$$

Proof. See Arbogast et al. (2022). □

1.4 Mixed Finite Elements

We then construct mixed finite elements from the introduced direct serendipity space. Now recall Ciarlet's definition Ciarlet (2002) of a finite element.

Definition 1.4. A triple $(E, X(E), \varphi_j)$ defines a finite element.

- Element $E \in \mathfrak{T}_h$;
- Space of shape function $X(E)$, and a set of shape function $\varphi_i \in X(E)$. The dimension of the shape function space $\dim(X(E)) = n$;
- A set of continuous and linear functionals ϕ_j

$$\phi_j : X(E) \supset \mathfrak{X}(E) \to \mathbb{R}, \tag{1.32}$$

such that unisolvence condition is achieved as

$$\langle \phi_j, \varphi_i \rangle = \delta_{ij}, \quad i, j \in \{1, 2, 3, \dots, n\} \tag{1.33}$$

In the following sections, we state our shape function space constructed using first and second order direct serendipity space and corresponding degrees of freedom. We also provide explicit construction of shape functions.

We introduce a H^1 and a $H(\mathrm{div})$ conforming mixed element space constructed with the pre-defined direct serendipity space for Darcy and Stokes problems.

1.4.1 A Darcy Model Problem

Consider a model problem for Darcy equation as

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{f}, \text{ in } \Omega \\ \nabla \cdot \mathbf{u} &= 0, \text{ in } \Omega \\ \mathbf{u} &= \mathbf{0}, \text{ on } \partial\Omega \end{aligned} \tag{1.34}$$

We seek solution in the following energy space as

$$\begin{aligned} \mathbb{V}_D &= \{\mathbf{v} \in (H(\mathrm{div}; \Omega))^2 : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \text{ on } \partial\Omega\}, \\ \mathbb{U}_D &= \{\mathbf{u} \in (H(\mathrm{div}; \Omega))^2 : \mathbf{u} \cdot \boldsymbol{\nu} = \mathbf{0} \text{ on } \partial\Omega\} = \mathbb{V}_D, \\ \mathbb{W}_D &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R}. \end{aligned} \tag{1.35}$$

We write weak form with these energy space, and consider the standard variational form : find $\mathbf{u} \in \mathbb{U}_D$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\mathbf{u}, \mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \ \forall \mathbf{v} \in \mathbb{V}_D, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad \forall \omega \in \mathbb{W}_D. \end{aligned} \tag{1.36}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{D,h} \subset \mathbb{V}$ and $\mathbb{W}_{D,h} \subset \mathbb{W}$, and consider the discretized problem: find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ such that

$$\begin{aligned} (\mathbf{u}_h, \mathbf{v}_h) - (p_h, \nabla \cdot \mathbf{v}_h) &= (\mathbf{f}, \mathbf{v}_h), \ \forall \mathbf{v}_h \in \mathbb{V}_{D,h}, \\ (\nabla \cdot \mathbf{u}_h, \omega_h) &= 0, \qquad \forall \omega_h \in \mathbb{W}_{D,h}. \end{aligned} \tag{1.37}$$

Now we construct a pair of discretized space $\mathbb{V}_{D,h} \times \mathbb{W}_{D,h}$ for this Darcy model problem in the next section.

1.4.2 A reduced $H(\mathbf{div})$ conforming space $\mathbb{V}_1^0$

We have some choice for $H(\text{div})$ confirming space. For example, we can choose the famous Ravier-Thomas element by Raviart and Thomas (1977). The serendipity space is the precursor of the Brezzi-Douglas-Marini element by Brezzi et al. (1985). We recover Arbogast-Correa element by Arbogast and Correa (2016) if we use mapped direct serendipity space. We generated a reduced $H(\text{div})$ approximation, that $\nabla \cdot \mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$.

We write the rotated 2D exact sequence for energy space as

$$H^1 \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2 \tag{1.38}$$

$$\cup \qquad\qquad \cup \qquad\qquad \cup \tag{1.39}$$

$$\mathcal{DS}_2 \xrightarrow{\text{curl}} \mathbf{V}_r^{r-1} \xrightarrow{\text{div}} \mathbb{P}_{r-1}. \tag{1.40}$$

We define the first order reduced $H(\text{div})$ conforming space with $\mathcal{DS}_2$ as

$$\mathbb{V}_1^0 = \mathbb{P}_1^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}}. \tag{1.41}$$

The reduced space $\mathbb{V}_1^0$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements. The dimension of the derived space is

$$\dim(\mathbb{V}_1^0) = 2 \times 3 + 2 = 8. \tag{1.42}$$

We need two independent basis functions on each edge of a quadrilateral E, one of which has a vanishing average flux on the edge and the other one has a non-vanishing average flux on the edge.

We state the degree of freedom of element as normal flux on each edge as

$$\int_{e_i} \mathbf{u} \cdot \boldsymbol{\nu} \varphi_i d\mathbf{x}, \quad \varphi_i \in \mathbb{V}_1^0. \tag{1.43}$$

We need two independent shape functions on each edge. We present a construction of two independent basis functions on each edge: $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{1.44}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{1.45}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i ds = 0$.

To begin with, we write curl of the auxiliary rational function R_i (1.16) as

$$\mathrm{curl} R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{1.46}$$

where the index i is understood with modulo of 4. We first present a construction of basis function $\boldsymbol{\varphi}_{l,i}$ as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \mathrm{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\mathrm{curl}R_i. \tag{1.47}$$

We then normalize it such that l_i varying from -1 to 1 as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.48}$$

where the index i is understood in the sense of modulo 4, and edge length $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing a edge, and show normal flux joining continuously as a linear function on the shared edge.

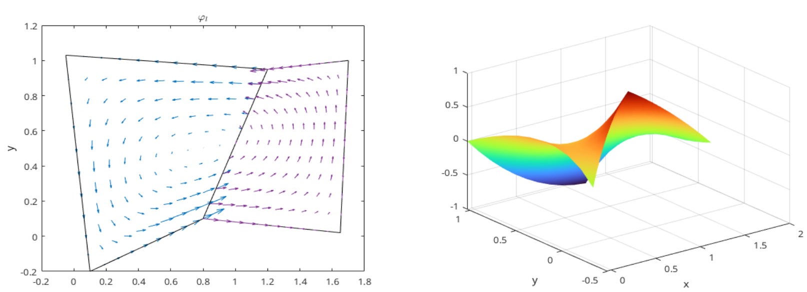


Figure 1.8: Linear basis function $\boldsymbol{\varphi}_l$.

The construction of a the constant basis function is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \big[\frac{1}{2}\varphi^2_{e,i} + \frac{1}{2}\varphi^2_{e,i+1}\big], \tag{1.49}$$

where the index i is understood in modulo of 4. Then define $\boldsymbol{\varphi}^*_{v,i}(\mathbf{x}) = \mathrm{curl}\phi^*_{v,i}$ so that

$$\boldsymbol{\varphi}^*_{v,i} \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{1.50}$$

We then define the "constant" basis function defined on the edge as

$$\boldsymbol{\varphi}_{c,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}^*_{v,i} + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{1.51}$$

where the index i is again understood in the sense of modulo of 4. The defined basis function $\boldsymbol{\varphi}_{c,i}$ has flux of constant value 1 on e_i and 0 on all the other edges. In Figure 1.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing a edge, and show normal flux joining continuously as a constant 1 on the shared edge.

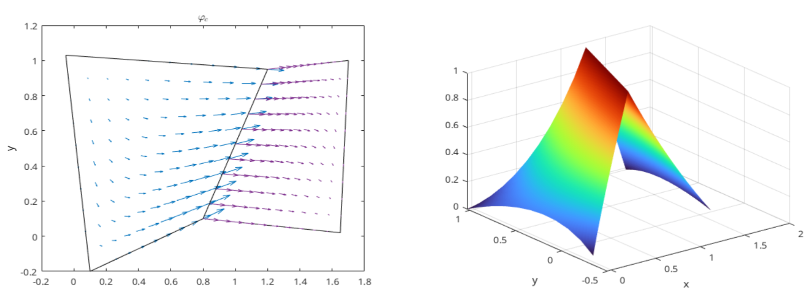


Figure 1.9: Constant basis function $\boldsymbol{\varphi}_c$.

1.4.3 Stability and Approximation Properties for $\mathbb{V}^0_1$

The accuracy and stability of such element has been well studied and tested in the work by Arbogast et al. (2022). We state the result directly for completeness. Consider a projection operatior π_D preserves element average divergence and average edge normal fluxes.

$$\pi_D : H(\mathrm{div};\Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}^0_1, \tag{1.52}$$

where $\epsilon > 0$. The global projection operator π is understood as combination of local operator π_E according to degree of freedom. The operator π also satisfies the commuting diagram property as

$$\mathcal{P}_{W_s} \nabla \cdot \mathbf{v} = \nabla \cdot \pi \mathbf{v}, \tag{1.53}$$

where $\mathcal{P}_{W_s}$ is the L^2-orthogonal projection operator onto $W_s = \nabla \cdot \mathbb{V}_1^0$, which is a constant value in our case.

Lemma 1.5. *Stability and approximation properties.*

Under the assumption of 1.3, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\|p - p_h\|_{m,\Omega} \leq Ch^{l+1-m} |p|_{l+1,\Omega}, \quad l = 0, 1, 2, \ldots, r, \tag{1.54}$$

where $p_h \in \mathcal{DS}_2$. We have

$$\begin{aligned} \|\mathbf{u} - \pi_D \mathbf{u}\|_{0,\Omega} &\leq C \|\mathbf{u}\|_{k,\Omega} h^k, \qquad k = 1, \ldots, 2 \\ \|\nabla \cdot (\mathbf{u} - \pi_D \mathbf{u})\|_{0,\Omega} &\leq C \|\nabla \mathbf{u}\|_{k,\Omega} h^k, \quad k = 0, \ldots, 1 \\ \|p - \mathcal{P}_{W_s} p\|_{0,\Omega} &\leq C \|p\|_{k,\Omega} h^k, \qquad k = 0, \ldots, 1 \end{aligned} \tag{1.55}$$

Moreover, the discrete inf-sup condition holds

$$\sup_{\mathbf{v}_h \in \mathbb{V}_1^0} \frac{(\omega_h, \nabla \cdot \mathbf{v}_h)}{\|\mathbf{v}_h\|_{H(div)}} \geq \gamma \|\omega_h\|_{0,\Omega}, \quad \forall \omega_h \in W_s, \tag{1.56}$$

holds for some $\gamma > 0$ independent of $\mathcal{T}_h$ and $h > 0$.

Proof. See Arbogast et al. (2022). □

1.4.4 A Stokes Model Problem

Consider a model problem for Stokes equation as Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{0} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.57}$$

We pick energy space for displacement and pressure as

$$\begin{aligned}
\mathbb{V}_S &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \ \text{ on } \ \partial\Omega\} = (H_0^1(\Omega))^2, \\
\mathbb{U}_S &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{0} \ \text{ on } \ \partial\Omega\} = \mathbb{V}_S, \\
\mathbb{W}_S &= \{p \in L^2(\Omega) : \int_\Omega p d\mathbf{x} = 0\} = L^2(\Omega)/\mathbb{R},
\end{aligned} \tag{1.58}$$

We write weak form with these energy space, and consider the standard variational form : find $\mathbf{u} \in \mathbb{U}_S$ and $p \in \mathbb{W}_S$ such that

$$\begin{aligned}
(\nabla\mathbf{u}, \nabla\mathbf{v}) - (p, \nabla\cdot\mathbf{v}) = (\mathbf{f}, \mathbf{v}),& \ \ \forall\mathbf{v} \in \mathbb{V}_S, \\
(\nabla\cdot\mathbf{u}, \omega) = 0,& \qquad \forall\omega \in \mathbb{W}_S.
\end{aligned} \tag{1.59}$$

We introduce two finite-dimensional subspace $\mathbb{V}_{S,h} \subset \mathbb{V}$ and $\mathbb{W}_{S,h} \subset \mathbb{W}$, and consider the discretized problem: find $(\mathbf{u}_h, p_h) \in \mathbb{V}_{S,h} \times \mathbb{W}_{S,h}$ such that

$$\begin{aligned}
(\nabla\mathbf{u}_h, \nabla\mathbf{v}_h) - (p_h, \nabla\cdot\mathbf{v}_h) = (\mathbf{f}, \mathbf{v}_h),& \ \ \forall\mathbf{v}_h \in \mathbb{V}_{S,h}, \\
(\nabla\cdot\mathbf{u}_h, \omega_h) = 0,& \qquad \forall\omega_h \in \mathbb{W}_{S,h}.
\end{aligned} \tag{1.60}$$

Now we construct a pair of discretized space $\mathbb{V}_{S,h}\times\mathbb{W}_{S,h}$ for this Stokes model problem in the next section.

1.4.5 Modified Bernardi-Raugel Element

Stable mixed finite element pair for Stokes problem has been widely researched. We are referring to the element discussed by Christine Bernardi (1985) and by Fortin (1981). A similar type of element was discussed by Arbogast and Wheeler (2005).

Inspired by the work in direct serendipity space, we consider the following modified space for each element $E \in \mathcal{T}_h$ as

$$V_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \operatorname{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \tag{1.61}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined on each edge e_i. One natural pick is the previously defined function $\varphi_{e,i}^{(2)} \in \mathcal{DS}_2$ as

$$\mathbf{b}_i = \varphi_{e,i}^{(2)}(\mathbf{x})\mathbf{n}_i, \tag{1.62}$$

which is quadratic when restricted to edge e_i according to equation (1.20). The quadratic bubble is guaranteed to be conforming since it evaluates zeros on two vertex and 1 at the middle point of the edge due to normalization. The integral of this quadratic bubble can be accurately computed by a three points gaussian quadrature rule.

Definition 1.5. For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following degrees of freedom:

(i) for corner point v_i and Cartesian direction unitnormal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF^{(\mathrm{i})}_{(v_i,\mathbf{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF^{(\mathrm{i})}_{(v_i,\mathbf{j})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF^{(ii)}_{e_i} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle,$$

We give an explicit construction of basis functions using driect serendipity space. We pick nodal basis on vertex v_i in two Cartesian direction as

$$N_{v_i,x} = \begin{bmatrix} \psi_{v,i}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \psi_{v,i}(\mathbf{x}) \end{bmatrix} \tag{1.63}$$

We pick edge basis on edge e_i as

$$N_e = \varphi^{(2)}_{e,i} \boldsymbol{\nu}_i \tag{1.64}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i, therefore bubble functions cannot be in the same space as nodal basis function. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function.

We define the corresponding discrete pressure sapce as

$$\mathbb{W}_0 = \{\omega \in L^2\Omega; \omega|_E \in P_0\} \cap L^2(\Omega)/\mathbb{R}, \tag{1.65}$$

which holds piece wise constant values for each element E. We provide stability and approximation analysis for this space pair $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ in the next section.

1.4.6 Stability and Approximation Properties for $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$

Before analyze the stability and approximation properties for the modified, we state some useful lemma.

Lemma 1.6. *Let K be a nondengenate convex quadrilateral. Assuming we have shape regularity 1.1. For each integer $m \geq 0$ and real number $p \in [1, \infty]$ there exist positive constants C_1, and C_2 independent of the geometry of E such that the following upper bound hold for all $v \in W_p^m(E)$:*

$$\begin{cases} |v|_{W_p^0(K)} \leq C_1 h_E^{2/p} |\tilde{v}|_{W_p^0(\tilde{K})}, \\ |v|_{W_p^1(K)} \leq C_2(\sigma^*, p) |\tilde{v}|_{W_p^1(\tilde{K})}, \end{cases} \tag{1.66}$$

where $\tilde{K}$ represents quadrilateral before scaling illustrated in Figure 1.1, and $\tilde{v}$ is function defined on $\tilde{K}$ before scaling.

Proof. See Girault and Raviart (2011). □

With the interpolation operator $\mathcal{I}_{\mathcal{DS}_1}$ defined in 1.27, we define the projection operator $\pi_h \in (H_0^1(\Omega))^2 \to \mathbb{V}_1^{BR} \cap (H_0^1(\Omega))^2$ by:

(i) For each corner point $v \in \partial E$

$$\pi_h \mathbf{v}(\mathbf{x}_v) = \mathcal{I}_{\mathcal{DS}_1} \mathbf{v}(\mathbf{x}_v). \tag{1.67}$$

(ii) On each edge $e \subset \partial E$,

$$\int_e (\pi_h \mathbf{v} - \mathbf{v}) \cdot \boldsymbol{\nu} ds = 0. \tag{1.68}$$

Lemma 1.7. *For all $\mathbf{v} \in (H^k(\Omega))^2, \quad k = 1, 2$. The operator π_h defined by* (1.67) *and* (1.68) *satisfies:*

$$(p_h, \nabla \cdot (\mathbf{v} - \pi_h \mathbf{v})) = 0, \quad \forall p_h \in \mathbb{W}_0 \tag{1.69}$$

Assuming shape regularity defined in 1.1, π_h should have the error bound as

$$\|\mathbf{v} - \pi_h \mathbf{v}\|_{1,\Omega} \leq Ch^m \|\mathbf{v}\|_{m+1,\Omega}, \quad m = 0, 1. \tag{1.70}$$

Proof. From the definition of degree of freedom (1.67) and (1.68), we decompose the projection operator π_h as

$$\pi_h \mathbf{v} = \mathcal{I}_{\mathcal{D}\mathcal{S}_1}\mathbf{v} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{1.71}$$

where the coefficient

$$\alpha_i = \Big(\int_{e_i} (\mathbf{v} - \mathcal{I}_{\mathcal{D}\mathcal{S}_1}\mathbf{v}) \cdot \boldsymbol{\nu}_i d\sigma \Big) / \int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i d\sigma. \tag{1.72}$$

We use the approximation property of interpolation operator in direct derendipity space in Lemma 1.4,

$$\|\mathbf{v} - \mathcal{I}_{\mathcal{D}\mathcal{S}_1}\mathbf{v}\|_{m,\Omega} \leq h_E^{2-m} |\mathbf{v}|_{2,E^*}, \quad m = 0, 1, \tag{1.73}$$

where $E^* = \Cup_{F \in \mathcal{T}_h, F \cap E \neq}$ defined in Lemma 1.3.

Lemma 1.6 indicates :

$$|\mathbf{b}_i|_{m,E} \leq C(\sigma^*) h_E^{1-m} |\tilde{\mathbf{b}}_i|_{1,\tilde{E}} \leq C(\sigma^*) h_E^{1-m}, \quad m = 0, 1. \tag{1.74}$$

We continue to estimate $|\alpha_i|$ as

$$\int_{e,i} \varphi_{e,i}^{(2)} ds = |e_i| \int_{\tilde{e}_i} \tilde{\varphi}_{e,i}^{(2)} d\tilde{s}. \tag{1.75}$$

According to Trace Theorem 1.1 the trace mapping is continuous, and Lemma 1.6, we estimate the term $|\alpha_i|$ as

$$|\alpha_i| \leq C \int_{\tilde{e},i} |\tilde{\mathbf{v}} - \tilde{\mathcal{I}}_{\mathcal{D}\mathcal{S}_1}\tilde{\mathbf{v}}| ds \leq C h_E^k h_E^{-1} |\mathbf{v}|_{k,E^*}. \tag{1.76}$$

The overall estimate is then

$$|\alpha_i| |\mathbf{b}_i|_{m,E} \leq C h^{k-m} |\mathbf{v}|_{k,E^*}. \tag{1.77}$$

Using the error estimation Lemma 1.4, and Bramble-Hilbert Lemma **??**, we get a global error estimate as

$$\|\mathbf{v} - \pi \mathbf{v}\|_{H^1(\Omega)} \leq C h |\mathbf{v}|_{H^2(\Omega)}. \tag{1.78}$$

$\square$

In addition to the approximability lemma 1.4, we also need stability result.

Lemma 1.8. *For any* $\mathbf{v} \in H_0^1(\Omega)$, *there exists* $\mathbf{v}_h \in \mathbb{V}_1^{\mathcal{BR}}$ *such that*

$$\|\mathbf{v}_h\|_{H^1(\Omega)} \leq C \|\mathbf{v}\|_{H^1(\Omega)} . \tag{1.79}$$

Proof. We express discrete solution $\mathbf{v}_h$ with the predefined projection operator $\mathbf{v}_h = \pi_h \mathbf{v}$, and denote the $\mathbf{w}_h = \mathcal{I}\mathbf{v}$, then we have the following inequality as

$$\|\mathbf{v}_h\| \leq \|\mathbf{w}_h\|_{H^1(\Omega)} + h_E^{-1}\Big\{ \|\mathbf{v} - \mathbf{w}_h\|_{L^2(\Omega)} + h|\mathbf{v} - \mathbf{w}_h|_{H^1(\Omega)}\Big\} \leq C \|\mathbf{v}\|_{H^1(\Omega)} . \tag{1.80}$$

□

Theorem 1.9. *With the established approximability Lemma 1.7, and the stability Lemma 1.8, for a convex domain* Ω, *there exists a constant* $\gamma > 0$ *such that*

$$\inf_{\omega \in \mathbb{W}_0} \sup_{\mathbf{v} \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}, \omega)}{\|\mathbf{v}\|_1 \|\omega\|_0} \geq \gamma > 0. \tag{1.81}$$

Proof.

$$\sup_{\mathbf{v} \in \mathbb{V}_1^{\mathcal{BR}}} \frac{(\nabla \cdot \mathbf{v}, \omega)}{\|\mathbf{v}\|_1 \|\omega\|_0} \geq \frac{(\nabla \cdot \pi_h \varphi, \omega)}{\|\pi_h \varphi\|_1 \|\omega\|_0} \geq \frac{\|\omega\|_0}{C \|\varphi_1\|} \geq \gamma > 0, \tag{1.82}$$

where π is a bounded projection operator. □

We conclude that our space $\mathbb{V}_1^{\mathcal{BR}} \times \mathbb{W}_0$ form a inf-sup stable pair with first order overall accuracy.

1.4.7 Numerical Test

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem but with a non-homogenous Dirichlet boundary condition:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega. \end{aligned} \tag{1.83}$$

We define a simple choice of energy space

$$\begin{aligned}\mathbb{V} &= \{\mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \ \text{ on } \partial\Omega\},\\ \mathbb{U} &= \{\mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g} \ \text{ on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V},\\ \mathbb{W} &= \{p \in L^2(\Omega)/\mathbb{R}\},\end{aligned} \tag{1.84}$$

where $\tilde{\mathbf{g}}$ is a finite energy lift. We assume that $\mathbf{g} \in (H^{1/2}(\Omega))^d$ and understand as a continuous extension $\tilde{\mathbf{g}} \in (H^1(\Omega))^d$ from the boundary into the domain. We write the problem in weak form as:

Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

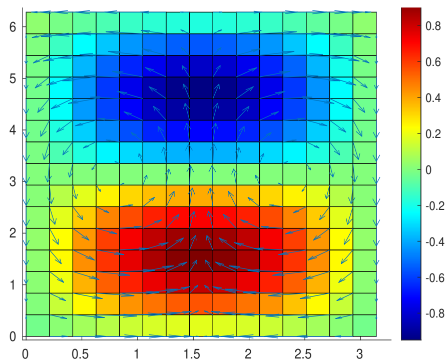
$$\begin{aligned}(\nabla\mathbf{u}, \nabla\mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \ \ \forall \mathbf{v} \in \mathbb{V},\\ (\nabla \cdot \mathbf{u}, \omega) &= 0, \qquad\quad \forall \omega \in \mathbb{W}.\end{aligned} \tag{1.85}$$

We consider the true solution of test problem (1.85) defined on a square region $\Omega = [0, \pi] \times [0, 2\pi]$. We manufacture a displacement and pressure as

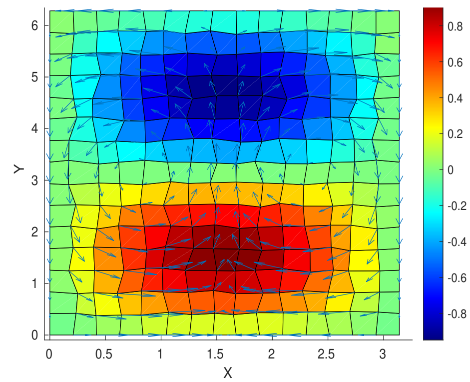
$$\mathbf{u} = \begin{bmatrix} \cos(x)\sin(y) \\ -\ \sin(x)\cos(y) \end{bmatrix}, \quad p = \sin(x)\sin(y), \tag{1.86}$$

which has 0 normal flux on the boundary. We deal with the constant kernal by enforcing a condition that $\int_\Omega p d\mathbf{x} = 0$ and solve the linear system directly. The numerical results are computed on rectangular 1.10a and logically rectangular quadrilateral 1.10b meshes. We calculate the L^2 norm of the residual (i.e., the difference) of the pressure, velocity, and derivative of velocity.

In Table 1.1 and Table 1.2 we display the result of a calculation of convergence on $n \times n$ rectangular and distorted grids, respectively, for $n = 8, 16, 32, 64$. Second order convergence rate is achieved by velocity and its corresponding derivatives. A first order convergence rate is achieved by pressure. Both meet our expectations for optimal convergence order.



(a) Rectangular Grids. Shown are pressure (color) and velocity (arrows).



(b) Distorted Grids. Shown are pressure (color) and velocity (arrows).

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.502E−1	1.052E−1	2.513E−2	6.130E−3
$L^2(\text{res}_p)$	1.039E+0	4.463E−1	2.093E−1	1.019E−1
$L^2(\text{res}_{du})$	2.632E+0	5.845E−1	1.374E−1	3.330E−2

Table 1.1: Rectangular Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Grid size	8×8	16×16	32×32	64×64
$L^2(\text{res}_u)$	4.603E−1	1.074E−2	2.567E−2	6.266E−3
$L^2(\text{res}_p)$	1.070E+0	4.670E−1	2.207E−1	1.078E−1
$L^2(\text{res}_{du})$	2.685E+0	5.981E−1	1.525E−1	4.580E−2

Table 1.2: Distorted Grids. L^2-error in the residual of the solution u (res_u), the pressure p (res_p), and the of derivatives of the solution Du (res_{du}).

Works Cited

Todd Arbogast and Jerry L. Bona. *Functional Analysis for the Applied Mathematician*. Chapman & Hall, 2025.

Todd Arbogast and Maicon R. Correa. Two families of h(div) mixed finite elements on quadrilaterals of minimal dimension. *SIAM Journal on Numerical Analysis*, 54(6):3332–3356, 2016. doi: 10.1137/15M1013705. URL `https://doi.org/10.1137/15M1013705`.

Todd Arbogast and Mary F. Wheeler. A family of rectangular mixed elements with a continuous flux for second order elliptic problems. *SIAM Journal on Numerical Analysis*, 42(5):1914–1931, 2005. doi: 10.1137/S0036142903435247. URL `https://doi.org/10.1137/S0036142903435247`.

Todd Arbogast, Zhen Tao, and Chuning Wang. Direct serendipity and mixed finite elements on convex quadrilaterals. *Numerische Mathematik*, 150:929–974, 2022.

J.H. Bramble and S. R. Hilbert. Estimation of linear functionals on sobolev spaces with application to fourier transform and spline interpolation. *SIAM J. Numer. Anal.*, 7(1):112–124, 1970. doi: 10.1137/0707006.

Franco Brezzi, Jim Douglas Jr., and L. D. Marini. Two families of mixed finite element for second order elliptic problems. *Numerische Mathematik*, 47:217–235, 1985.

Genevieve Raugel Christine Bernardi. Analysis of some finite elements for the stokes problem. *Math. of Comput.*, 44:71–79, 1985.

Philippe G. Ciarlet. *The Finite Element Method for Elliptic Problems*. Society for Industrial and Applied Mathematics, 2002. doi: 10.1137/1.9780898719208. URL `https://epubs.siam.org/doi/abs/10.1137/1.9780898719208`.

Todd Dupont and Ridgway Scott. Polynomial approximation of functions in sobolev spaces. *Mathematics of Computation*, 34(150):441–463, 1980. ISSN 00255718, 10886842. URL `http://www.jstor.org/stable/2006095`.

Michel Fortin. Old and new finite elements for incompressible flows. *International Journal for Numerical Methods in Fluids*, 1:347–364, 1981. URL `https://api.semanticscholar.org/CorpusID:122770101`.

Vivette Girault and Pierre-Arnaud Raviart. *Finite Element Methods for Navier-Stokes Equations: Theory and Algorithms*. Springer Publishing Company, Incorporated, 1st edition, 2011. ISBN 3642648886.

P. A. Raviart and J. M. Thomas. A mixed finite element method for 2-nd order elliptic problems. In Ilio Galligani and Enrico Magenes, editors, *Mathematical Aspects of Finite Element Methods*, pages 292–315, Berlin, Heidelberg, 1977. Springer Berlin Heidelberg. ISBN 978-3-540-37158-8.